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Pore-Scale Experimental Study of Particle Dynamics and Filter Cake Formation in Porous Media During Produced Water Reinjection

Tongtong Yu, Energy & Technology, NORCE Norwegian Research Centre AS, Bergen, Norway; Na Liu, Department of Physics and Technology, University of Bergen, Bergen, Norway; Ketil Djurhuus, Energy & Technology, NORCE Norwegian Research Centre AS, Bergen, Norway

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Abstract

Reinjecting produced water into reservoirs offers an environmentally friendly approach to water management, but the suspended impurities -- particularly solid particles -- could cause formation damage and decrease injectivity. This study investigates particle transport and retention in porous media by injecting synthetic produced water through a heterogeneous microfluidic chip -- which mimics reservoir pore structure -- under different conditions. Real-time visualization across sections with identical pore structure reveals that higher flow rates and smaller particle sizes enhance penetration, while larger particles and lower rates promote early retention near the inlet. Particle accumulation is closely linked to local flow conditions and pore geometry: narrow pore throats discourage settling due to higher velocities, while wider ones promote retention. The early stage of internal filter cake formation is observed near the inlet, and pore-scale heterogeneity disperses particles away from the main flow path, resulting in a broader spatial distribution. These findings highlight the complex interplay between hydrodynamic forces and pore structure in governing particle behavior, offering insights into optimizing water management strategies and minimizing formation damage in porous media.

Introduction

Produced water reinjection is an effective method for managing production water with zero harmful discharge. However, impurities such as oil droplets, solid particles, and microbial cells present in the produced water can lead to formation damage and injectivity loss (Kurbasov & Feilberg, 2025) (Hsi, Dudzik, Lane, Buettner, & Neira, 1994) (Obe, Fashanu, Idialu, Akintola, & Abhulimen, 2017). Among these, solid particles are considered one of the primary contributors to permeability reduction and are the focus of this study. Transported with the injected water, a single particle can deposit in porous media through mechanisms such as size exclusion, gravitational settling, and adsorption, depending on particle size. Additionally, particle bridging, as well as progressive clogging may occur when multiple particles accumulate at pore constrictions (Gao, 2007) (Xia, et al., 2022) (Okaybi, Roman, & Soullaine, 2025).

Particle migration, deposition, and aggregation in porous media constitute a complex process influenced by multiple factors, including properties of the porous media, particles characteristics, fluid dynamics, and the forces acting between them (McDowell-Boyer, Hunt, & Sitar, 1986) (Liu, Cui, Zhang, & Huang, 2015). These processes accelerate reservoir aging and result in unrecoverable oil. A deeper understanding of particle penetration and filter cake formation mechanisms -- particularly under varying operational conditions -- can inform the development of improved water management process, enable prediction of injectivity variations, and support proactive mitigation measures, thereby promoting sustainable reservoir development.

Many experimental and analytical studies have investigated permeability variations and particle migration and aggregation in porous media under various conditions (Hammadi, Ahfir, Alem, & Wang, 2017) (Vetter, Kandarpa, Stratton, & Veith, 1987) (Mirshekari, Dadvar, Modarress, & Dabir, 2013) (Sacramento, et al., 2015) (Li, et al., 2025) (Kokubun, et al., 2019) (Feia, et al., 2015) (Yu, Standnes, Kowalewski, Carlsen, & Djurhuus, 2025). A subset of these studies has focused on microscale phenomena. For example, Levaché et al. demonstrated that elevated temperatures can reduce particle clogging in the presence of oil due to the lowered oil viscosity (Levaché, et al., 2023). A novel clogging mechanism -- referred to as dendrite clogging -- was identified by Okaybi et al., describing clogging initiated from a single deposition site that branches out (Okaybi, Roman, & Soulaïne, 2025). Kusaka et al. examined the influence of Péclet numbers on deposition morphology around cylindrical obstacles, highlighting the role of flow conditions in shaping particle accumulation patterns (Kusaka, Duval, & Adachi, 2010).

The fundamental mechanisms of particle transport in porous media are not yet fully understood, despite their importance in governing particle penetration, filter cake formation, and the resulting variations in injectivity during water injection in reservoirs. This study employs microfluidic technology to investigate the mechanisms underlying particle transport and deposition in porous media, which remains less commonly studied. By designing microfluidic devices that replicate the heterogeneous pore structures of real reservoirs, the study enables precise control over experimental conditions and real-time visualization of particle behavior. This approach facilitates a detailed examination of the interplay between hydrodynamic forces and pore-scale interactions, thereby enhancing the understanding of particle accumulation mechanisms.

Synthetic produced water -- brine with suspended particles -- is injected into a reservoir-like microfluidic chip (also called micromodel in the context) under high pressure to examine particle transport and retention. The effects of injection rate, particle size, and pore structure on filter cake formation and injectivity reduction are systematically evaluated. By analyzing particle behavior within a consistent pore structure at varying distances from the inlet, as well as transitions between pore throats and pore bodies, this study provides valuable insights into the relationship between injection rate, transport distance, and in situ flow dynamics through real-time visualization.

Materials and methods

Experimental setup

The experimental setup is illustrated in Fig. 1. A silicon microfluidic chip features four ports located at each corner. Two open channels with the width of 200 μm connect the ports: one channel runs between port 1 to port 2, and the other between port 3 to port 4. The micromodel, measuring 27×22 mm with a depth of 30 μm , comprises 36 repetitions of a heterogeneous pattern, which was designed based on the structure of a natural sandstone. The pore network has a total porosity of 61% and a pore volume of 11 μL . The microfluidic chip surface is strongly hydrophilic, with a surface roughness of approximately 100 nm. Although the porosity of 61% is higher than that of typical subsurface reservoirs, we employ this microfluidic pore network with such high porosity to better visualize and isolate the fundamental processes governing particle transport in porous media.

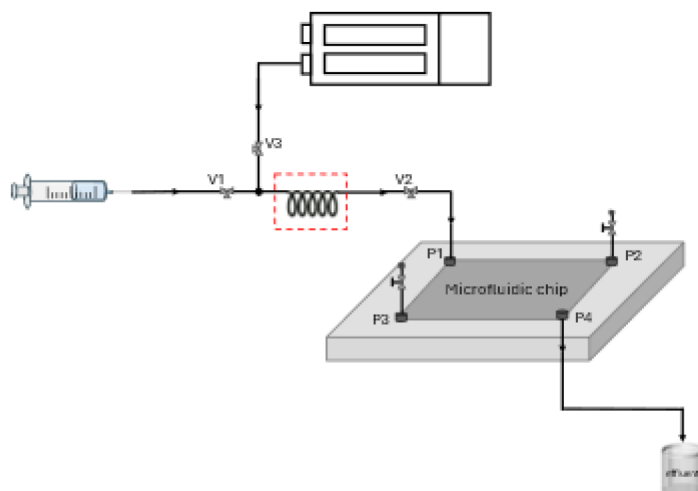


Figure 1—Schematic illustration of the experimental setup. Port 1 of the microfluidic chip serves as the inlet, while port 4 as the outlet, port 2 and 3 are sealed during operation. Prepared solution is first injected into the coil tubing (highlighted by the red dashed square) using a syringe with valve V1 open and valves V2 and V3 closed. The solution is then introduced into the microfluidic chip via a precision pump, with valves V2 and V3 open and valve V1 closed. Throughout the experiment, the microfluidic chip is placed horizontally and maintained at a constant temperature of 30 °C.

This microfluidic chip is mounted on a chip holder positioned under a Zeiss microscope (Axio Zoom V16, Zeiss) with a moving stage, illuminated by a cold light source (CL 9000 LED). The experimental setup enabled us to study the entire porous network area, yielding 1-2-1 unique images of different parts of the pore space with a resolution of 1.1 $\mu\text{m}/\text{pixel}$.

The system pressure was maintained at 10 barg using a back pressure regulator (EB1ZF1 Equilibar Zero Flow) connected to a pressurized 300 mL N_2 cylinder. Port 1 served as the injection inlet and was connected to a 1 mL section of tubing pre-filled with the test solution. The solution was introduced into the chip using a high-precision Quizix Q5000 plunger pump. Port 4, located diagonally opposite the inlet and serving as the outlet, was connected to the back pressure regulator to ensure stable system pressure.

Solution preparation

A 3 wt% NaCl solution containing suspended quartz particles (BCR®-070, particle size range 1.20-20.00 μm) at a concentration of 70 ppmw (parts per million by weight) was used as an analogue for produced water in the experiments. To prevent particle aggregation and enhance solution stability, 50 ppmw of the nonionic surfactant Triton™X-405 was added.

Experimental procedure

This study comprises two individual experiments, both conducted using the same microfluidic setup and prepared solution, under a pore pressure of 10 barg and at a temperature of 30 °C.

Experiment 1: Nine vertically aligned sections of the micromodel from inlet to outlet -- each featuring the same heterogeneous pattern -- were selected for observation, as illustrated in Fig. 2. During the solution injection, port 1 was used as the inlet and port 4 as the outlet. Sequential images were captured across all nine sections in continuous loops, with each section imaged approximately every 13 seconds. This process was repeated 60 times. Four injection rates were applied: 1.0, 1.5, 2.0, 2.5 $\mu\text{L}/\text{min}$. The microfluidic chip has an effective flow area of 0.403 mm^2 and the selected injection rates implies a frontal advancement of 4-9 m/d, which is representative of near-well flow for an offshore injection well with injection rates on the order of 1000-5000 m^3/d (Bhullar, Karlsen, Owe, Seland, & Tran, 1999), (Awan, Teigland, & Kleppe, 2008), (Aase, Bjørkum, & Nadeau, 1996), (Denholm, 1987)

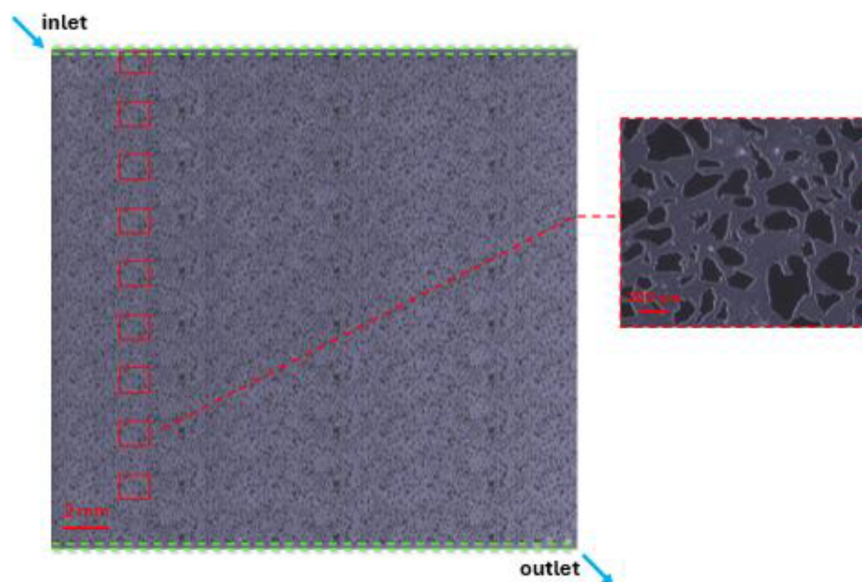


Figure 2—Microfluidic chip used in Experiment 1, with port 1 (top left) serving as the inlet and port 4 (bottom right) as the outlet. Nine sections, marked by red squares and aligned vertically, were selected for analysis. The sections share identical pore structure but vary in distance from the inlet, allowing for investigations of spatial effects on particle transport. Green dashed lines indicate two flow channels: one connecting port 1 to port 2, and another connecting port 3 to port 4. A magnified view of one selected region is shown on the right, highlighting the pore geometry.

This experiment investigated particle transport, deposition, and accumulation with a focus on:

- i. the transition between pore throat and pore bodies,
- ii. behavior within identical pore structures at varying distances from the inlet, and
- iii. the influence of different injection rates on these processes.

Experiment 2: The entire micromodel was examined to study particle behavior at a larger spatial scale. Images were acquired sequentially from the top-left to the bottom-right of the micromodel, following a s-shaped scanning pattern during particle injection at a constant flow rate of $0.5 \mu\text{L}/\text{min}$, with fluid introduced through port 3 and exiting through port 2. Each complete scan of the micromodel was completed in approximately 600 seconds.

This experiment focused on analyzing the temporal evolution of particle transport and spatial distribution throughout the full pore network.

Image processing and analysis

The image data generated from both experiments were converted from .czi to .png format using ZEISS ZEN Microscopy Software to enable further analysis. Subsequent image processing and data handling were carried out in Python (v3.11) using the *OpenCV*, *NumPy*, *pandas*, and *Patchify* libraries.

The .png files from both experiments were subjected to a three-step image segmentation process to isolate particles, grains, and grain walls. The original RGB images (Fig. 3a) were first converted to grayscale for threshold-based segmentation.

- i. **High-threshold segmentation:** Grayscale images were thresholded using a high-intensity value to distinguish brighter features, such as particles and grain walls, from the darker grains and pore spaces. The resulting binary image highlighted these features in white against a black background (Fig. 3b).
- ii. **Low-threshold segmentation:** The same grayscale images were thresholded using a lower intensity value to isolate darker grains from the brighter pore spaces. This produced a binary image where

grains appeared white on a black background, enabling the identification of grain boundaries, which are marked in red in Fig. 3c.

- iii. **Particle isolation:** To extract particles alone, the grain wall regions identified in step (ii) were subtracted from the high-threshold image (Fig. 3b). The resulting image displayed only the particles on a black background. These segmented particles were then overlaid with the grain regions to visualize grains containing injected particles (Fig. 3d).

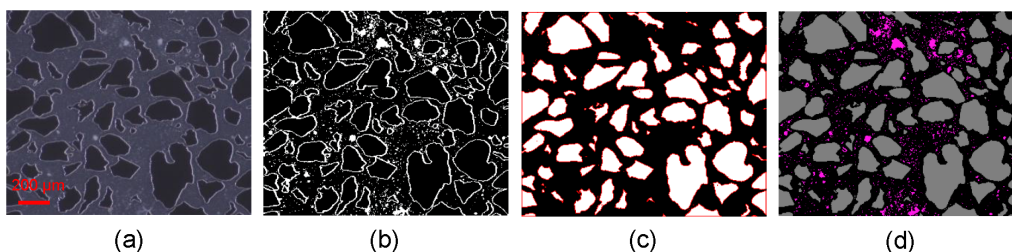


Figure 3—Illustration of the image segmentation process. (a): Original RGB image converted from .czi file (black regions corresponding to silicon grains and white regions representing particles settled in the blue pore space). (b): Binary image highlighting particles and grain walls in white. (c): Binary image showing grains in white on a black background, with grain walls boundaries overlaid in red for visualization. (d): Final segmentation result with particles shown in pink and grains in gray.

The accuracy of the segmentation process was evaluated through manual inspection of selected frames, with an estimated image analysis error of approximately 7%.

Results and discussions

Particle penetration depth

During Experiment 1, particle movement was directly observed through the microscope. In sections located further from the injection inlet, particle velocity gradually decreased, and beyond a certain distance, no particle motion was detected. Additionally, the extent of particle transport was influenced by the injection rate: higher injection rates resulted in particles penetrating deeper into the pore network of the micromodel. Fig. 4a presents the relationship between injection rate and the maximum distance from the inlet at which visibly moving particles (size > 2 μm) were observed. A linear correlation was found between the injection rate and the maximum particle transport distance, with a correlation coefficient of 0.98.

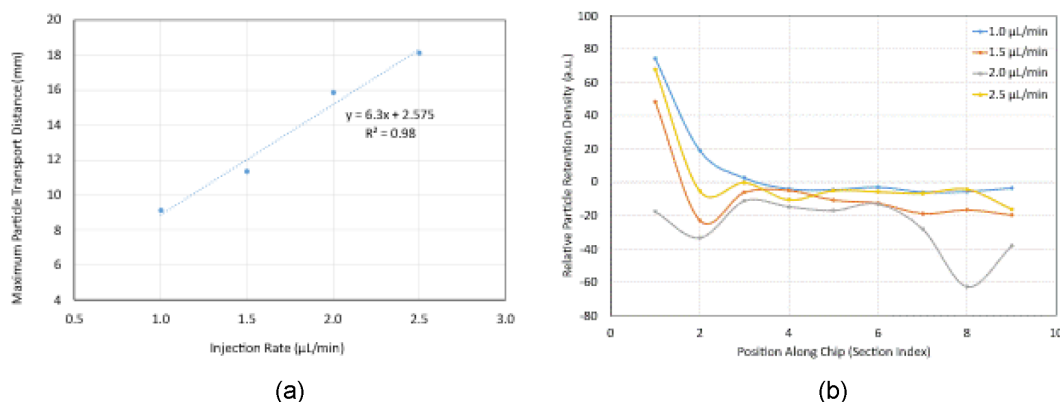


Figure 4—(a): Relationship between injection rate and maximum particle transport distance from the inlet, as determined by visual observation. A linear trend was observed, with the fitted equation $y = 6.3x + 2.575$ and a high correlation coefficient ($R^2 = 0.98$). (b): Relationship between relative particle retention density and the nine positions along the micromodel. Comparison is made across four different injection rates: 1.0, 1.5, 2.0, 2.5 $\mu\text{L}/\text{min}$.

Particle accumulation density in each section was quantified from the image segmentation process. The pixel number corresponding to particles was determined from the segmented images, and the difference between the final and initial particle counts was divided by the image size to calculate a particle retention density, which is plotted on the y-axis in Fig. 4b. The x-axis represents the position along the flow path, with smaller values corresponding to locations closer to the injection inlet. Particle density profiles were obtained at selected sections along the micromodel (as defined in Fig. 2) for four injection rates from 1.0 to 2.5 $\mu\text{L}/\text{min}$.

Given that the structural configuration is identical across all nine sections, the observed variations in particle retention are attributed to two primary factors: (1) the distance from the injection inlet and (2) the injection rate. At all injection rates, most particles accumulated within the first section of the micromodel, indicating that the region closest to the inlet acts as a primary retention zone, consistent with previous findings (Liu, Cui, Zhang, & Huang, 2015). This suggests that a significant portion of particles is retained near the inlet area during injection. Among the tested conditions, the highest particle retention density is observed at the lowest injection rate of 1.0 $\mu\text{L}/\text{min}$, suggesting that slower flow enhances particle retention within the micromodel.

The remaining particles -- typically smaller in size -- continue to migrate with the current are transported beyond the primary retention zone, eventually reaching the rear sections of the porous medium (Liu, Cui, Zhang, & Huang, 2015). This migration is influenced by the flow strength, which in this study is controlled by the injection rate (Meigel, Darwent, Bastin, Goehring, & Alim, 2022). Higher injection rates exert stronger drag forces on the particles, as described by Eq. 1. The enhanced drag forces enable particles to overcome pore-scale barriers -- such as adhesion to pore walls, constrictions at narrow pore throats, and early entrapment along the flow path -- thereby promoting deeper penetration into the micromodel (Das, Mukherjee, & Muralidhar, 2017). This effect is evident in the experimental results: the highest particle density within the primary retention zones is observed at the lowest injection rate of 1.0 $\mu\text{L}/\text{min}$, extending up to the third section. At injection rate of 2.5 $\mu\text{L}/\text{min}$, particle density remains relatively stable in the rear zones, suggesting reduced early retention, a more uniform distribution along the flow path, and potentially deeper penetration beyond section 9. For the particle size range used in this experiment (1.20 - 20.00 μm), final aggregation is governed by a combination of external forces (e.g., gravity and hydrodynamic effects), electrokinetic mechanisms (such as surface charge effects), and local interactions among trapped particles within the porous structure (McDowell-Boyer, Hunt, & Sitar, 1986) (Liu, Cui, Zhang, & Huang, 2015).

$$F_D = \frac{1}{2} C_D \rho A v^2 \quad \text{Eq. 1}$$

Where F_D is the drag force, C_D is the drag coefficient, ρ is the fluid density, A is the cross-sectional area, and v is the fluid velocity.

However, unlike the penetration length, which increases more consistently with flow rate (Fig. 4a), the effect of flow rate on particle retention is non-linear within the tested range. As the injection rate increases, particle accumulation near the inlet decreases, likely due to higher shear forces dislodging previously retained particles. At 2.0 $\mu\text{L}/\text{min}$, a negative particle density was recorded after 15 image capture loops. This suggests that particles were either flushed out of the pore space or became more compacted within the depth of the microfluidic chip. The latter possibility -- particle compaction in the out-of-plane direction -- cannot be confirmed with the current 2D imaging setup and warrants further investigation.

Effect of pore geometry on particle transport

Particle transport in porous media is strongly influenced by the geometry of the pore network. Variations in pore size, connectivity and local constrictions give rise to heterogeneous velocity fields, which in turn control where particles are likely to migrate, accumulated or be retained. Particles tend to settle preferentially in low-velocity regions (Fig. 5), while high velocity zones promote continued transport or even re-entrainment of previously deposited particles. Portion of the particles is retained through mechanisms such

as straining and physiochemical interactions between the particles and the porous matrix (McDowell-Boyer, Hunt, & Sitar, 1986).

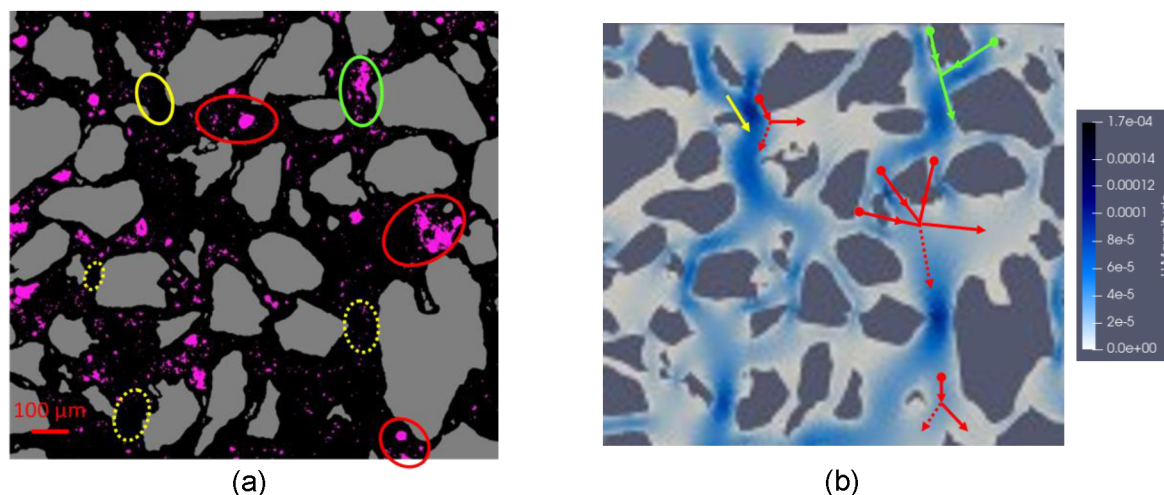


Figure 5—Comparison between experimental and simulated flow behavior in the microfluidic chip. (a) particle distribution obtained through image segmentation of microfluidics experiments at a flow rate of 0.5 $\mu\text{L}/\text{min}$ (section 2, $t=15$). Yellow circles (solid and dashed) indicate regions of particle absence corresponding to high-velocity zones; red circles highlight particle accumulation in low-velocity zones; and the green circle marks particle retention influenced by factors other than flow rate. (b) Simulated velocity magnitude in the pore network, with the color bar displayed on the right. The velocity field represents the steady-state flow velocity magnitude (refer to <https://github.com/daavid00/pymm.git> for details on the simulation setup).

Fig. 5 compares experimental and simulated data to illustrate how pore geometry and local flow patterns affect particle retention. Fig. 5a presents the particle distribution obtained via image segmentation, where particles are highlighted in pink and grains in gray. Fig. 5b displays the velocity magnitude field derived from numerical simulation, using a predefined convergence criterion. In Fig. 5b, the regions of dark blue indicate zones of higher fluid velocity within the porous medium -- the darker the blue, the faster the flow. These connected blue regions highlight the preferential flow paths and help trace how particles might be transported or trapped along the path. A comparison of Fig. 5a and 5b reveals a strong spatial correlation between particle accumulation and variation in flow velocity, highlighting the role of geometry-driven velocity variation in governing particle transport behavior.

In the region with the highest flow rate, indicated by the yellow arrow in Fig. 5b, particles are notably absent in Fig. 5a (marked by a solid yellow circle). Similar patterns are observed in other regions highlighted by dashed yellow circles in Fig. 5a, suggesting that particles tend not to settle in high-velocity, geometrically constrained zones.

Pore branching also plays a significant role in directing particle transport. The red arrows in Fig. 5b illustrate a case where a mainstream (indicated by an arrow with a starting dot) branches into multiple tributaries. The narrower tributary (dashed arrow) exhibiting higher velocity, shows little to no particle accumulation, while the wider tributary (solid arrow), with lower velocity, contains a noticeable buildup of particles (red circles in Fig. 5a). This behavior is consistent with preferential settling in slower flow paths, typically associated with larger pore cross-sections and reduced shear forces.

However, particle accumulation and distribution are influenced by multiple factors beyond flow rate, which is primarily governed by pore throat width. For example, the region marked by a green circle in Fig. 5a exhibits dense particle accumulation, despite the corresponding area in Fig. 5b (indicated by green arrows) showing a high simulated flow velocity. Based on the simulation, particles would be expected to pass through the pore throat rather than being trapped. The pore structure reveals that this section of the flow path involves two tributaries (marked by arrows with starting dots) converging into a single main channel. Although the main channel has a wider diameter than either tributary -- suggesting a lower velocity from

geometric considerations -- the convergence of flows increases the local fluid volume and thus the flow velocity. Despite the high flow velocity in this region, particles are observed to settle, suggesting that factors such as adhesion, electrostatic interactions, or surface roughness of the porous matrix may play a significant role under this condition (Bagrezaie, Dabir, & Rashidi, 2022). Additionally, the pore structure at the junction may induce flow disturbances, such as eddies or low-pressure zones, which facilitate particle deposition even in areas of elevated velocity (Hunt, Wray, & Moin, 1988). The combined effects of physicochemical interactions and localized flow disturbances appear to outweigh the influence of hydrodynamic forces, thereby becoming the dominant mechanisms driving particle retention under realistic scenarios.

Particle settlement in the pore network

Fig. 6 illustrates the spatial distribution of particle accumulation across the micromodel over time. Over the course of the experiment, particle injections into the micromodel ceased after several hours -- no new particles were observed entering the pore network. This observation suggests either the formation of an external filter cake upstream of the micromodel (outside the field of view) or particle settling within the injection pipeline over time. Consequently, particle accumulation within the micromodel reflects primarily early-stage deposition and transport behavior, rather than continuous input.

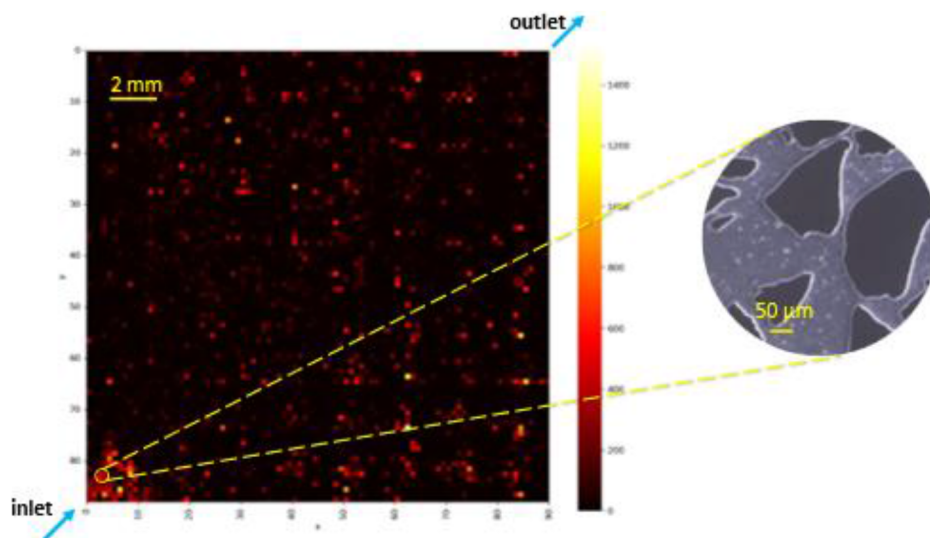


Figure 6—Accumulated particle spatial distribution across the microfluidic chip. The heatmap represents particle coverage area (μm^2), with darker regions indicating lower accumulation and brighter regions indicating higher accumulation. Axes are scaled in increments of 200 pixels. Injection was performed from port 3 (bottom left, inlet) to port 2 (top right, outlet). A magnified inset highlights accumulated particles in porous media close to inlet at the end of the experiment.

Following the image segmentation process described in Fig. 3, the increase in particles was quantified. For each 200×200 -pixel region, the number of newly accumulated particles was calculated and converted into coverage area (μm^2). This data is visualized as a heatmap in Fig. 6, illustrating the spatial distribution of accumulated particles across the entire micromodel. The x- and y-axis values represent increments of 200 pixels, corresponding to the grid of analyzed regions. The injection was performed from port 3 (bottom left), and the outlet was located at port 2 (top right).

The injected particles accumulated predominantly in the bottom-left corner of the micromodel, near the inlet at port 3, approximately 2.2 mm into the porous medium along the diagonal direction. This region corresponds to the primary retention zone identified in Experiment 1. As shown in the magnified circle in Fig. 6, particles are densely concentrated in this area, however, the pore spaces are not fully blocked. This indicates the early stage of internal filter cake formation, driven by progressive clogging mechanism.

While a portion of the particles were not retained near the inlet, many migrated throughout the micromodel. Their distribution in downstream regions appears more spatially uniform and does not strictly follow the diagonal flow path from inlet to outlet. This suggests that after a short initial phase of migration through the porous medium, particle movement is governed more by the geometry and connectivity of the pore structure than by the bulk flow direction. The topology and heterogeneity of the porous medium is playing a critical role in directing fluid flow and particle behavior. As a result, particles are distributed more broadly across the micromodel, leading to a more spatial uniform distribution compared to the initial, flow-aligned migration pattern, consistent with previous simulation-based findings in heterogeneous porous media (Zhang, et al., 2021).

Conclusions

Particle transport in porous media is strongly influenced by injection rate, particle size, pore geometry, and local flow conditions. Accumulation patterns reflect a complex interplay of structural heterogeneity and hydrodynamic forces, shaping particle mobility and distribution throughout the micromodel.

Higher injection rate causes deeper particle penetration in porous media by enhancing drag forces that help particles overcome pore-scale retention mechanisms. In contrast, at lower injection rates, particles are more likely to be retained in the initial sections of the micromodel, forming distinct early retention zones. Additionally, larger particles exhibit shallower invasion depths compared to smaller one. These patterns highlight the critical role of flow strength in determining how far particles can travel through the porous structure. The observed linear relationship between injection rate and transport distance confirms that flow rate is a key driver of particle mobility.

Particle accumulation is closely linked to local flow conditions and pore geometry. Narrow pore throats typically generate higher velocities, which discourage particle settling, while wider regions with lower flow rates tend to promote accumulation. However, deviations from this trend -- such as particle buildup in high-velocity zones -- suggest that additional factors like flow convergence and divergence, as well as complex pore structures, appear to play a significant role in directing particle transport and retention. These observations highlight that particle behavior results from a complex interplay of hydrodynamic forces and local structural effects, underscoring the value of combining experimental and simulation approaches for a more comprehensive understanding.

When observing the entire micromodel, initial particle accumulation occurs near the inlet due to progressive clogging, making the onset of internal filter cake formation. This retention zone is located close to the injection point and is characterized by dense particle buildup. Beyond this region, the pore-scale heterogeneity of the medium plays a critical role in influencing particle transport. It disperses particles away from the original flow path, resulting in a broader and more spatially uniform distribution across the micromodel. This behavior highlights the importance of pore structure in governing particle migration beyond the initial accumulation zone.

In real cases, solid particles present in the produced water can deposit within porous media and form filter cakes, particularly in the near-well regions. Higher injection rates can push particles deeper into the formation, reducing clogging in the near-well region and mitigating the rise in injection pressure when same amount particles are introduced. However, higher injection rates also increase the risk of particle deposition over a broader region, potentially impairing permeability further into the formation. Therefore, injection strategies must be carefully tailored to balance transport efficiency with the risk of formation damage, depending on specific production goals. Particle size also plays a critical role -- larger particles tend to penetrate less and accelerate clogging in near-well region. As such, filtration of produced water should be optimized based on the pore throat size of the target reservoir to balance filtration cost, production efficiency, and the need for post-clogging mitigation. Beyond these considerations, combining experimental

observations with numerical simulations offers a powerful approach for strategy development and enable better prediction of particle behavior under varying reservoir conditions.

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References

- Awan, A., Teigland, R., & Kleppe, J. (2008, June). A Survey of North Sea Enhanced-Oil-Recovery Projects Initiated During the Years 1975 to 2005. *SPE Reservoir Evaluation & Engineering*, **11**(03), 497–512. doi:<https://doi.org/10.2118/99546-PA>
- Bagrezaie, M., Dabir, B., & Rashidi, F. (2022, September). A novel approach for pore-scale study of fines migration mechanism in porous media. *Journal of Petroleum Science and Engineering*, 216. doi:<https://doi.org/10.1016/j.petrol.2022.110761>
- Bhullar, A., Karlsen, D., Owe, K., Seland, R., & Tran, K. (1999, November). Dating reservoir filling — a case history from the North SeaDiamer. *Marine and Petroleum Geology*, **16**(7). doi:[https://doi.org/10.1016/S0264-8172\(99\)00028-8](https://doi.org/10.1016/S0264-8172(99)00028-8)
- Das, M., Mukherjee, P., & Muralidhar, K. (2017). Equations Governing Flow and Transport in Porous Media. Springer, Cham. doi:https://doi.org/10.1007/978-3-319-69866-3_2
- Denholm, J. (1987, December). North Sea Development Drilling, South East Forties: A Case Study. *SPE Drilling Engineering*, **2**(04), 387–392. doi:<https://doi.org/10.2118/15448-PA>
- Feia, S., Dupla, J., Ghabezloo, S., Sulem, J., Canou, J., Onaisi, A., ... Aubry, E. (2015, September). Experimental investigation of particle suspension injection and permeability impairment in porous media. *Geomechanics for Energy and the Environment*, **3**, 24–39. doi:<https://doi.org/10.1016/j.gete.2015.07.001>
- Gao, C. (2007). Factors Affecting Particle Retention in Porous Media. *Emirates Journal for Engineering Research*, **12**(3), 1–7.
- Hammadi, A., Ahfir, N.-D., Alem, A., & Wang, H. (2017, March). Effects of Particle Size Non-Uniformity on Transport and Retention in Saturated Porous Media. *Transport in Porous Media*, **118**, 85–98. doi:<https://doi.org/10.1007/s11242-017-0848-6>
- Hsi, C., Dudzik, D., Lane, R., Buettner, J., & Neira, R. (1994). Formation Injectivity Damage Due to Produced Water Reinjection. SPE Formation Damage Control Symposium, Lafayette, Louisiana. Lafayette, Louisiana: The Society of Petroleum Engineers. doi:<https://doi.org/10.2118/27395-MS>
- Hunt, J., Wray, A., & Moin, P. (1988). Eddies, streams, and convergence zones in turbulent flows. Studying Turbulence Using Numerical Simulation Databases, 2. Proceedings of the 1988 Summer Program. California. Retrieved from <https://ntrs.nasa.gov/citations/19890015184>
- Kokubun, M., Muntean, A., Radu, F., Kumar, K., Pop, I., Keilegavlen, E., & Spildo, K. (2019, November). A pore-scale study of transport of inertial particles by water in porous media. *Chemical Engineering Science*, **207**, 397–409. doi:<https://doi.org/10.1016/j.ces.2019.06.036>
- Kurbasov, M., & Feilberg, K. (2025, February). Permeability shifts in chalk core during produced water reinjection. *Geoenergy Science and Engineering*, 245. doi:<https://doi.org/10.1016/j.geoen.2024.213471>
- Kusaka, Y., Duval, J., & Adachi, Y. (2010). Morphology and Breaking of Latex Particle Deposits at a Cylindrical Collector in a Microfluidic Chamber. *Environmental Science & Technology*, **44**(24), 9413–9418. doi:<https://doi.org/10.1021/es1026689>
- Levaché, B., Santos de Pera, N., Levaché, B., Questel, M., Panizza, P., Lequeux, F., Passade-Boupat, N. (2023, January). Jams and Cakes: A Closer Look on Well Clogging Mechanisms in Microscale Produced Water ReInjection Experiments. *Transport in Porous Media*, **147**, 143–156. doi:<https://doi.org/10.1007/s11242-023-01900-0>
- Li, H., Gao, G., Hao, M., Tian, R., Lu, Z., & Zhao, Y. (2025). Simulation on detachment and migration behaviors of mineral particles induced by fluid flow in porous media based on CFD-DEM. *Geoenergy Science and Engineering*, 247. doi:<https://doi.org/10.1016/j.geoen.2025.213727>
- Liu, Q., Cui, X., Zhang, C., & Huang, S. (2015, October). Experimental investigation of suspended particles transport through porous media: particle and grain size effect. *Environmental Technology*, **37**(7), 854–864. doi:<https://doi.org/10.1080/09593330.2015.1088578>
- McDowell-Boyer, L., Hunt, J., & Sitar, N. (1986). Particle Transport Through Porous Media. *Water Resources Research*, **22**(13), 1901–1921. doi:<https://doi.org/10.1029/WR022i013p01901>

- Meigel, F., Darwent, T., Bastin, L., Goehring, L., & Alim, K. (2022). Dispersive transport dynamics in porous media emerge from local correlations. *Nature Communications*, **13**(1). doi:<https://doi.org/10.1038/s41467-022-33485-5>
- Mirshekari, B., Dadvar, M., Modarress, H., & Dabir, B. (2013). Modelling and simulation of multiphase flow formation damage by fine migration including the multilayer deposition effect. *International Journal of Oil, Gas and Coal Technology*, **6**(6). doi:<http://10.1504/ijogct.2013.056711>
- Obe, I., Fashanu, T., Idialu, P., Akintola, T., & Abhulimen, K. (2017). Produced water re-injection in a non-fresh water aquifer with geochemical reaction, hydrodynamic molecular dispersion and adsorption kinetics controlling: model development and numerical simulation. *Applied Water Science*, **7**, 1169–1189. doi:<https://doi.org/10.1007/s13201-016-0490-4>
- Okaybi, W., Roman, S., & Soulaire, C. (2025). Progressive colloidal clogging mechanism by dendritic build-up in porous media. *Soft Matter*, **21**(28), 5687–5698. doi:<https://doi.org/10.1039/d5sm00285k>
- Sacramento, R., Yang, Y., You, Z., Waldmann, A., Martins, A., Vaz, A., ... Bedrikovetsky, P. (2015, February). Deep bed and cake filtration of two-size particle suspension in porous media. *Journal of Petroleum Science and Engineering*, **126**, 201–210. doi:<https://doi.org/10.1016/j.petrol.2014.12.001>
- Vetter, O., Kandarpa, V., Stratton, M., & Veith, E. (1987). Particle Invasion Into Porous Medium and Related Injectivity Problems. SPE International Symposium on Oilfield Chemistry. San Antonio, Texas: The Society of Petroleum Engineers. doi:<https://doi.org/10.2118/16255-MS>
- Xia, T., Feng, Q., Wang, S., Shu, Q., Zhang, Y., & Sun, Y. (2022, July). A Numerical Study of Particle Migration in Porous Media During Produced Water Reinjection. *Journal of Energy Resources Technology*, **144**(7). doi:<https://doi.org/10.1115/1.4052165>
- Yu, T., Standnes, D., Kowalewski, E., Carlsen, D., & Djurhuus, K. (2025, August). Darcy-Scale Experimental Investigation of Intermittent Particle Transport in Porous Media and its Impact on Injectivity. *PREPRINT (Version 1) available at Research Square*. doi:<https://doi.org/10.21203/rs.3.rs-7452899/v1>
- Zhang, C., Kaito, K., Hu, Y., Patmonoaji, A., Matsushita, S., & Suekane, T. (2021, March). Influence of stagnant zones on solute transport in heterogeneous porous media at the pore scale. *Physics of Fluids*, **33**(3). doi:<https://doi.org/10.1063/5.0038133>
- Aase, N., Bjørkum, P., & Nadeau, P. (1996). The Effect of Grain-Coating Microquartz on Preservation of Reservoir Porosity. *American Association of Petroleum Geologists Bulletin (AAPG Bulletin)*, **80**(10), 1654–1673. doi:<https://doi.org/10.1306/64EDA0F0-1724-11D7-8645000102C1865D>